

Netflix Movies and TV Shows Data Cleaning Report – Python

Summary:

This project focuses on cleaning the Netflix Movies and TV Shows dataset sourced from **Kaggle**. The goal was to handle missing values, fix data types, and prepare the dataset for further analysis.

1. Imported the necessary libraries

```
#Importing the necessary libraries

import pandas as pd
import warnings
warnings.filterwarnings('ignore')
```

2. Loaded the dataset

```
#Loading the dataset
df = pd.read_csv('Netflix_Movies_and_TV_Shows.csv')
df.head()
```

✓ 0.1s  Open 'df' in Data Wrangler Python

	show_id	type	title	director	cast	country	date_added
0	s1	TV Show	3%	Missing value	João Miguel, Bianca Comparato, Mic	Brazil	August 14, 2020
1	s2	Movie	7:19	Jorge Michel Grau	Demian Bichir, Héctor Bonilla, Oscar	Mexico	December 23, 2016
2	s3	Movie	23:59	Gilbert Chan	Tedd Chan, Stella Chung, Henley Hui	Singapore	December 20, 2018
3	s4	Movie	9	Shane Acker	Elijah Wood, John C. Reilly, Jennifer C	United States	November 16, 2017
4	s5	Movie	21	Robert Luketic	Jim Sturgess, Kevin Spacey, Kate Bos	United States	January 1, 2020

3. Information of the dataset

```
df.info()
✓ 0.0s

<class 'pandas.DataFrame'>
RangeIndex: 7787 entries, 0 to 7786
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   show_id         7787 non-null   str
1   type            7787 non-null   str
2   title           7787 non-null   str
3   director        5398 non-null   str
4   cast            7069 non-null   str
5   country         7280 non-null   str
6   date_added      7777 non-null   str
7   release_year    7787 non-null   int64
8   rating          7780 non-null   str
9   duration        7787 non-null   str
10  listed_in       7787 non-null   str
11  description      7787 non-null   str
dtypes: int64(1), str(11)
memory usage: 3.4 MB
```

4. Total Rows and Columns in the dataset

```
# Rows and Columns

print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])
```

✓ 0.0s

Number of Rows: 7787
Number of Columns: 12

5. Checked for duplicates

```
# Checking for duplicates

print(df.duplicated().sum())
```

✓ 0.0s

0

There were no duplicate rows in this dataset.

6. Checked for missing values

```
# Check for missing values

df.isnull().sum()
```

[41] ✓ 0.0s

...

	# 0
show_id	0
type	0
title	0
director	2389
cast	718
country	507
date_added	10
release_year	0
rating	7
duration	0

7. Handled missing values

```
# Handling missing values
# For 'Director', 'Cast', 'Country', 'Rating' we can fill missing values with 'Unknown'

df['director'] = df['director'].fillna('Unknown', inplace=True)
df['cast'] = df['cast'].fillna('Unknown', inplace=True)
df['country'] = df['country'].fillna('Unknown', inplace=True)
df['rating'] = df['rating'].fillna('Unknown', inplace=True)

print(df.isnull().sum())
```

✓ 0.0s

```
show_id      0
type         0
title        0
director     0
cast         0
country      0
date_added   10
release_year  0
rating       0
duration     0
listed_in    0
description  0
dtype: int64
```

7. Extracted the year from 'date_added' and created a new column 'year_added'

```
# Extracting the year from 'date_added' and creating a new column 'year_added'

df['year_added'] = df['date_added'].str.extract(r'(\d{4})')
df['year_added']
```

✓ 0.0s

	year_added
0	2020
1	2016
2	2018
3	2017
4	2020
5	2017
6	2020
7	2019
8	2019
9	2017

8. Checked for nulls and filled with mode.

```
df['year_added'].isnull().sum()
✓ 0.1s
np.int64(10)

# Fill nulls in year_added with mode

mode_year = df['year_added'].mode()[0]
df['year_added'] = df['year_added'].fillna(mode_year)
print(df['year_added'].isnull().sum())
✓ 0.0s
0
```

9. Converted year_added column to Int64 dtype for further smooth analysis.

```
# Converting 'year_added' to integer type

df['year_added'] = df['year_added'].astype('Int64')
print(df['year_added'].head())
✓ 0.0s
0    2020
1    2016
2    2018
3    2017
4    2020
Name: year_added, dtype: Int64
```

10. Filled nulls with Mode and converted the data type from string to datetime.

```
# Fill nulls in date_added with mode

mode_date = df['date_added'].mode()[0]
df['date_added'] = df['date_added'].fillna(mode_date)
print(df['date_added'].isnull().sum())
✓ 0.0s
0

# Converting 'date_added' to datetime type

df['date_added'] = pd.to_datetime(df['date_added'], errors='coerce')
print(df['date_added'].dtype)
✓ 0.0s
datetime64[us]
```

11. Removed leading and trailing spaces from string columns using Lambda Function.

```
# Removing leading and trailing spaces from string columns

df = df.apply(lambda x: x.str.strip() if x.dtype == 'object' else x)
✓ 0.0s
```

12. Renamed the 'listed_in' column to 'genre' for better understanding.

```
# Renaming the 'listed_in' column to 'genre'

df.rename(columns={'listed_in': 'genre'}, inplace=True)
print(df.columns)
✓ 0.0s

Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added',
      'release_year', 'rating', 'duration', 'genre', 'description',
      'year_added'],
      dtype='str')
```

13. Removed '#' symbol from the 'title' column

```
# Removing '#' from the 'title' column

df['title'] = df['title'].str.replace('#', '', regex=False)
✓ 0.0s
```

14. Saving the cleaned dataset to a new CSV file

```
df.to_csv('Netflix_Cleaned.csv', index=False)
print("Saved!")
✓ 0.2s

Saved!
```

Conclusion:

The dataset was successfully cleaned, null values were handled, datatypes were corrected and the cleaned file was exported as Netflix_Cleaned.csv